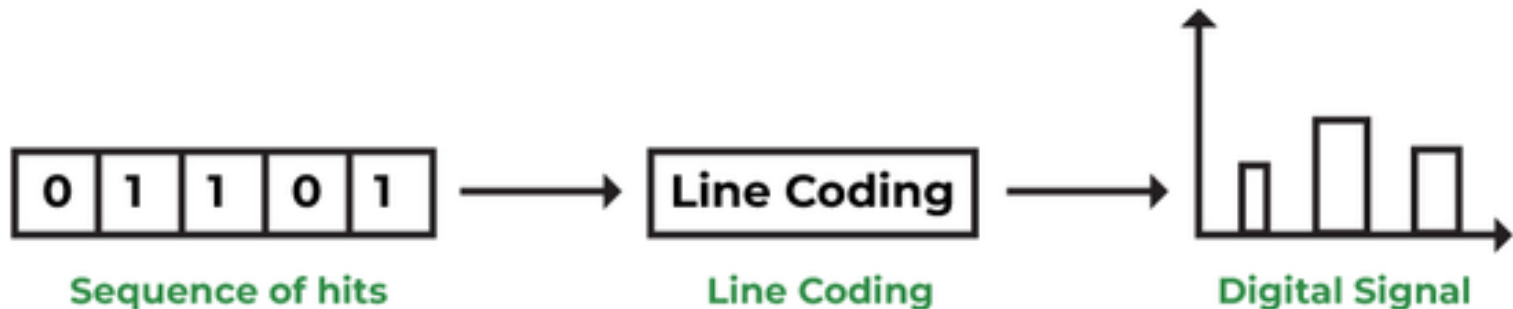


LINE CODING

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Line Coding

- Line coding is the process of converting digital data (0s and 1s) into digital signals for transmission over a physical medium.



Need of Line Coding

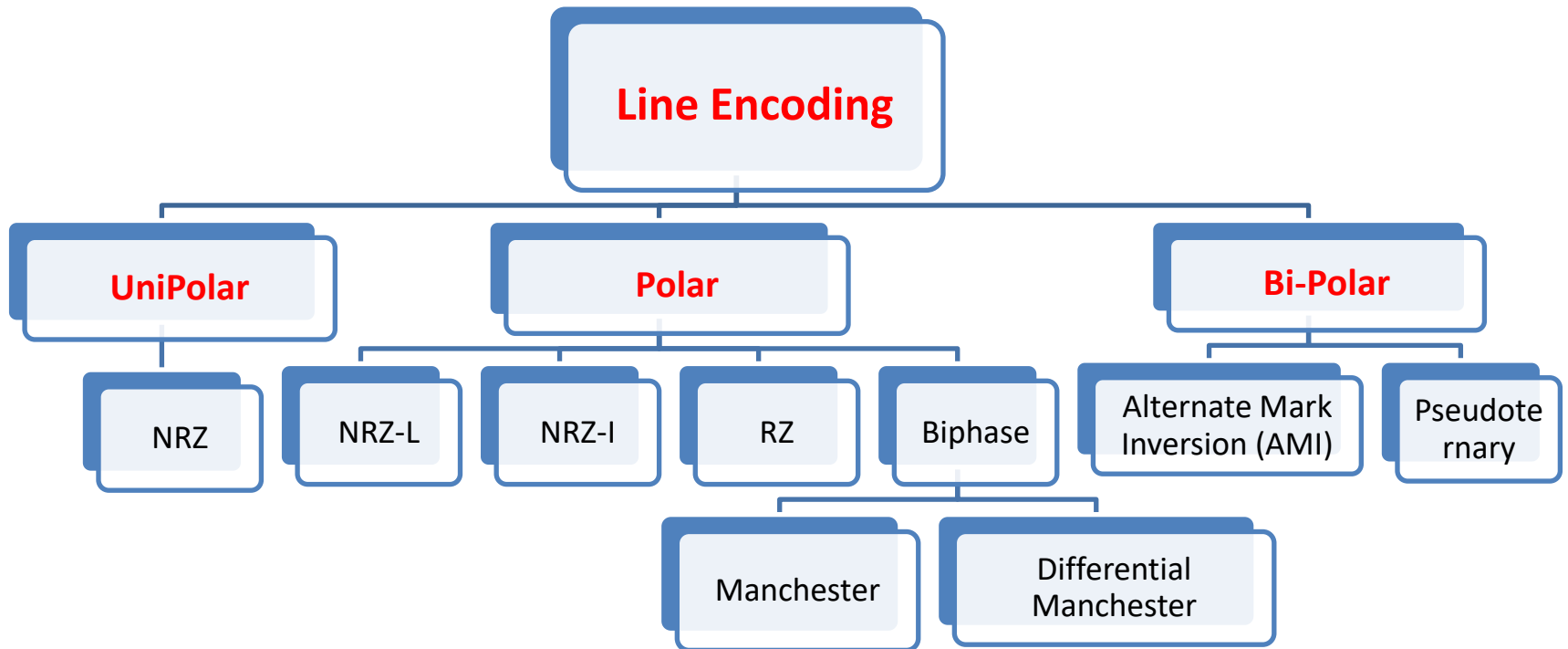
- Physical medium understands signals, not bits
- Synchronization between sender and receiver
- Reduce noise effect
- Avoid DC component and distortion

(DC component is the non-zero average value of a signal voltage over time.)

Why is DC Component a Problem?

- Transformer-based systems cannot pass DC signals
- Causes baseline wandering at the receiver
- Increases signal distortion
- Creates problems in long-distance transmission

Types of Line Encoding



UniPolar Line Coding

- Signal uses **only one polarity** of voltage
- Voltage is either **+V** or **0**
- Signal stays on **one side of the voltage axis**

Unipolar NRZ(Non Return to Zero)

Rule:

- $1 \rightarrow +V$
- $0 \rightarrow 0$

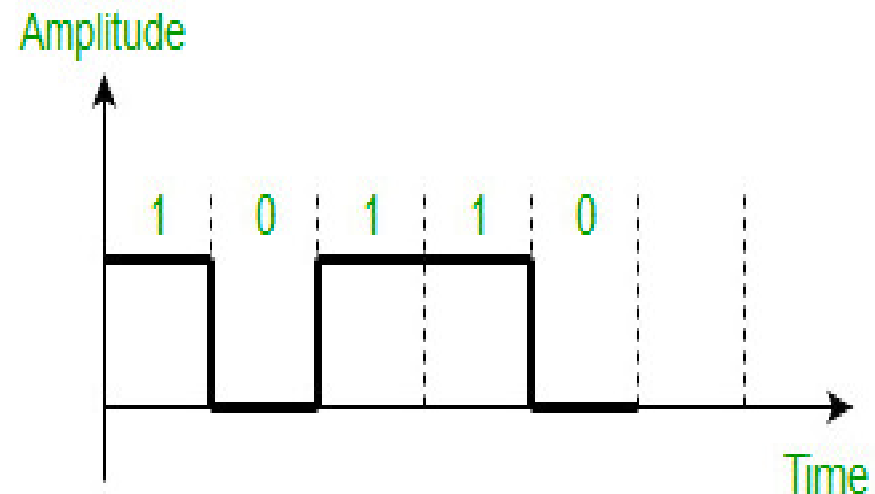
Advantage:

- Simple implementation

Drawbacks:

- DC component
- Poor synchronization

For example: Data = 10110.



Signal does not return to zero at the middle of the bit thus it is called NRZ.

Unipolar Line Coding

Advantages:

- Very simple to design and implement
- Requires less circuitry

Disadvantages:

- High DC component present
- Poor synchronization
- Not suitable for long-distance transmission

Polar Line Coding

- **Why “Polar”?**
“Polar” refers to **two opposite polarities**.
- Signal uses **both +V and -V**
- Voltage swings on **both sides of the axis**
- That is why it is called **Polar** line coding.

Polar NRZ-L(NRZ-Level)

Level of the voltage determines the value of the bit

Rule:

- $1 \rightarrow +V$
- $0 \rightarrow -V$

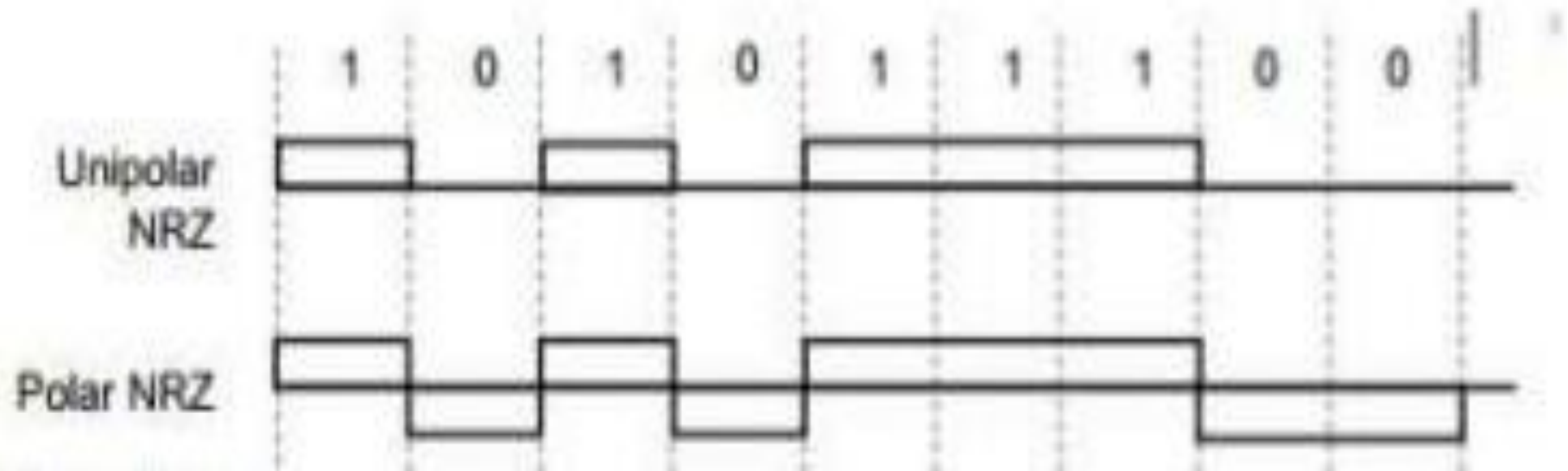
Advantages:

- Better than unipolar

Drawbacks:

- Long same bits cause sync problem

EXAMPLE: Polar NRZ



Polar NRZ-I(Non-Return to Zero-Invert)

Rule:

- 1 → Invert signal
- 0 → No change

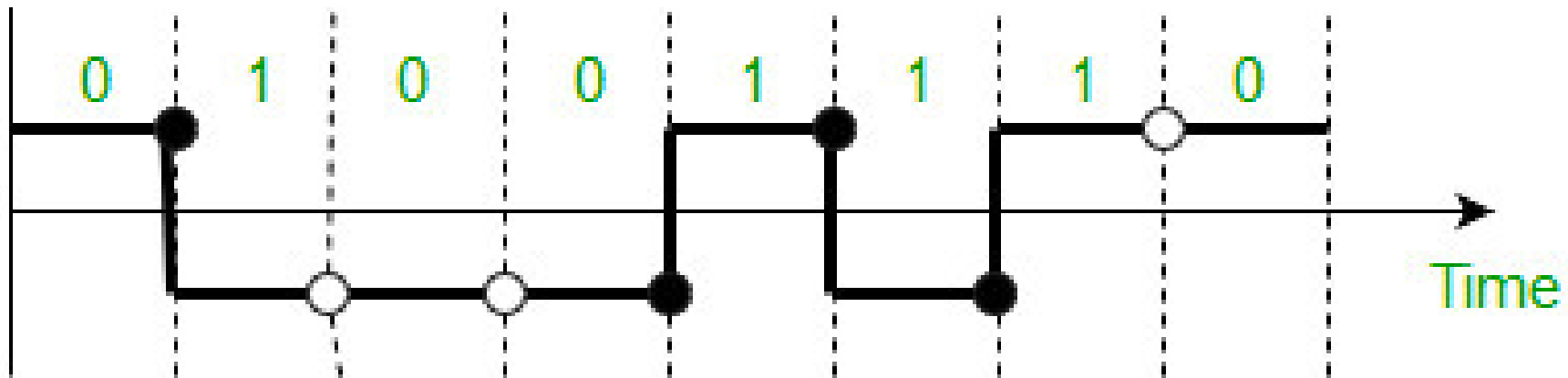
Advantages:

- Better synchronization

Drawbacks:

- Long 0s problematic

Example: Data = 01001110.



NRZ-I

Polar RZ(Return to Zero)

In RZ line coding, the signal **returns to zero voltage in the middle of each bit interval.**

Rule:

- $1 \rightarrow +V$ then 0
- $0 \rightarrow -V$ then 0

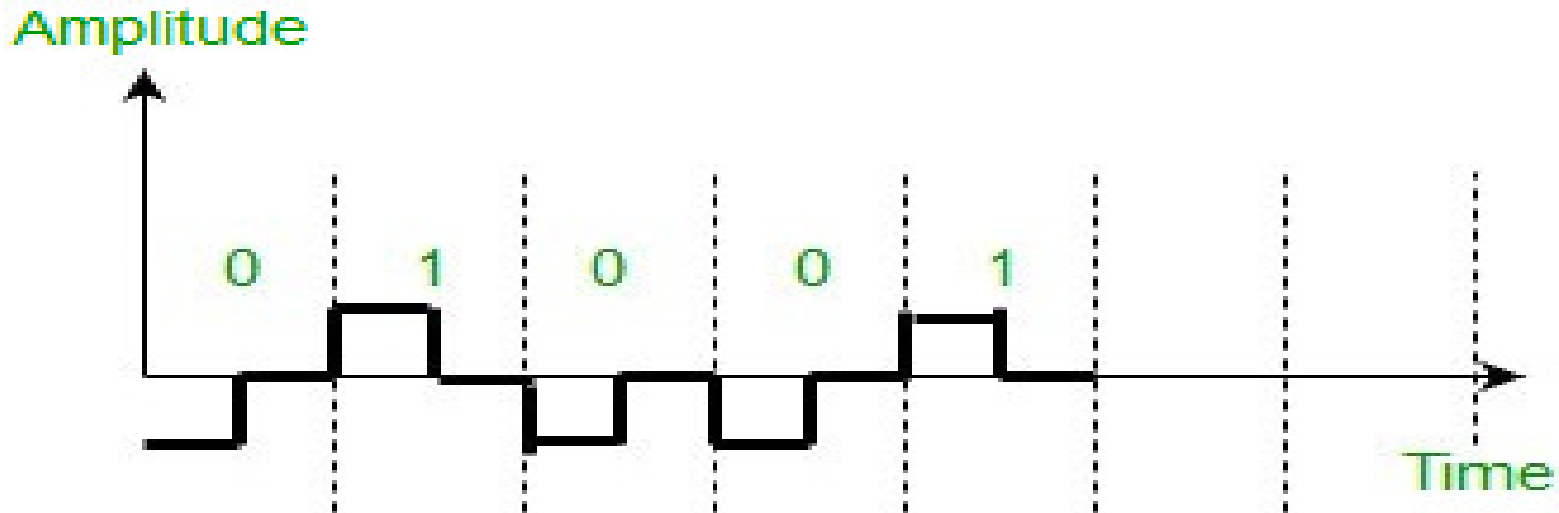
Advantages:

- Better synchronization

Drawbacks:

- High bandwidth
- Complexity as it uses three levels of voltage

Example: Data = 01001.



The logic we are using here to represent data is that for bit 1 half of the signal is represented by $+V$ and half by zero voltage and for bit 0 half of the signal is represented by $-V$ and half by zero voltage.

Manchester Encoding

Combination of the RZ (transition at the middle of the bit) and NRZ-L schemes.

The duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half.

Rule:

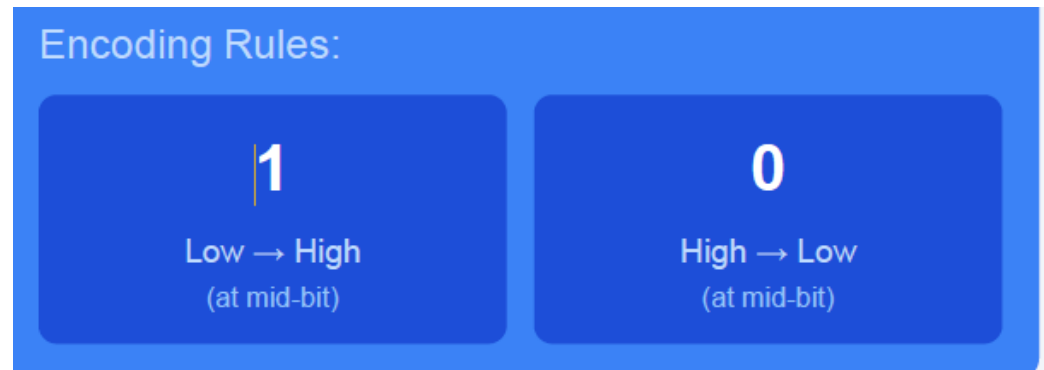
- 0 → High to Low
- 1 → Low to High

Advantages:

- Excellent synchronization
- No DC component

Drawbacks:

- Double bandwidth



Differential Manchester

Combination of the RZ (transition at the middle of the bit) and NRZ-I schemes.

- There is always a transition at the middle of the bit but the bit values are determined at the beginning of the bit.
- If the next bit is 0, there is a transition, if the next bit is 1, there is no transition.

Rule:

- **Mid-bit transition always**
- **0 → Start transition**
- **1 → No start transition**

Encoding Rules:

0

Transition at
beginning of bit

1

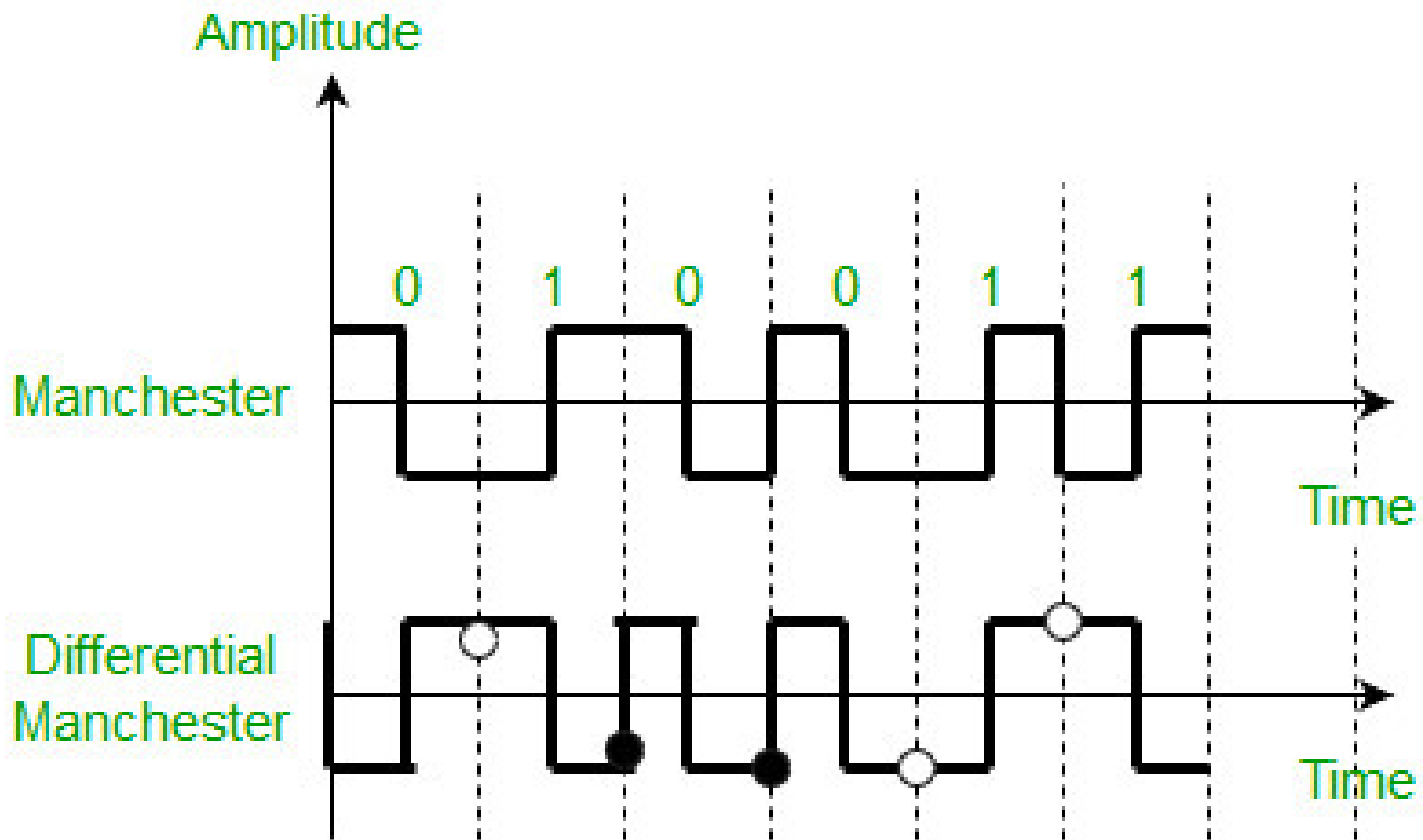
No transition at
beginning of bit

Advantages:

- Noise resistant

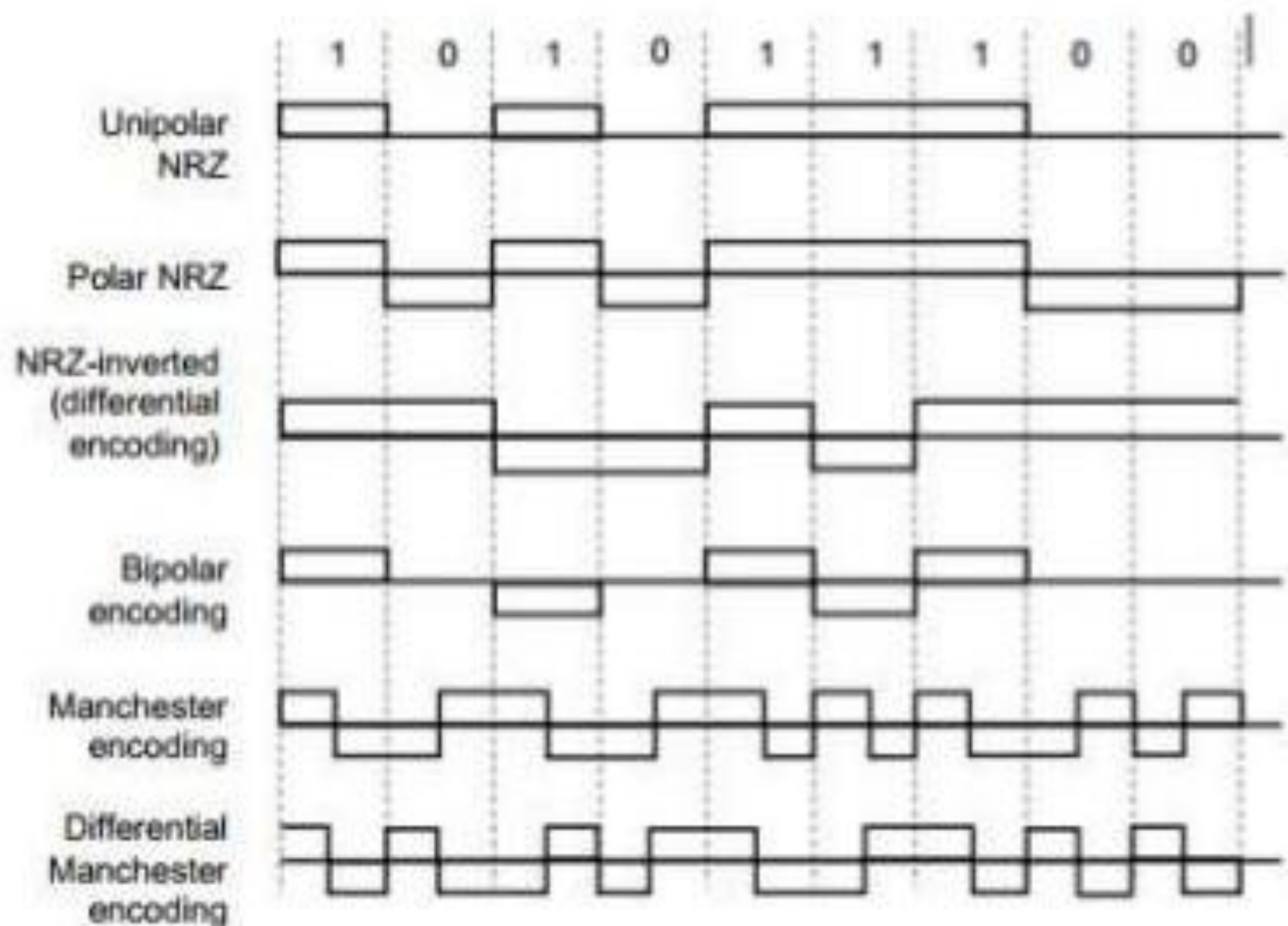
Drawbacks:

- Complex, high bandwidth



- No Inversion
- Inversion

Types of Line Coding



Polar Line Coding

Advantages:

- Better noise immunity than unipolar
- DC component is reduced
- Uses both positive and negative voltages

Disadvantages:

- Synchronization problem for long sequences of same bits
- More complex than unipolar

Bipolar Line Coding

- **Why “Bipolar”?**
“Bi” means **two**, but here it refers to **two polarities used alternately**.
- Three voltage levels are used: **+V, 0, -V**
- One bit uses **both positive and negative voltages alternately**
- That is why it is called **Bipolar** line coding.

Bipolar: Alternate Mark Inversion(AMI)

Rule:

- $0 \rightarrow 0$
- $1 \rightarrow +V, -V$ alternately

Advantages:

- No DC component

Drawbacks:

- Long zero sequence issue

Pseudoternary

Rule:

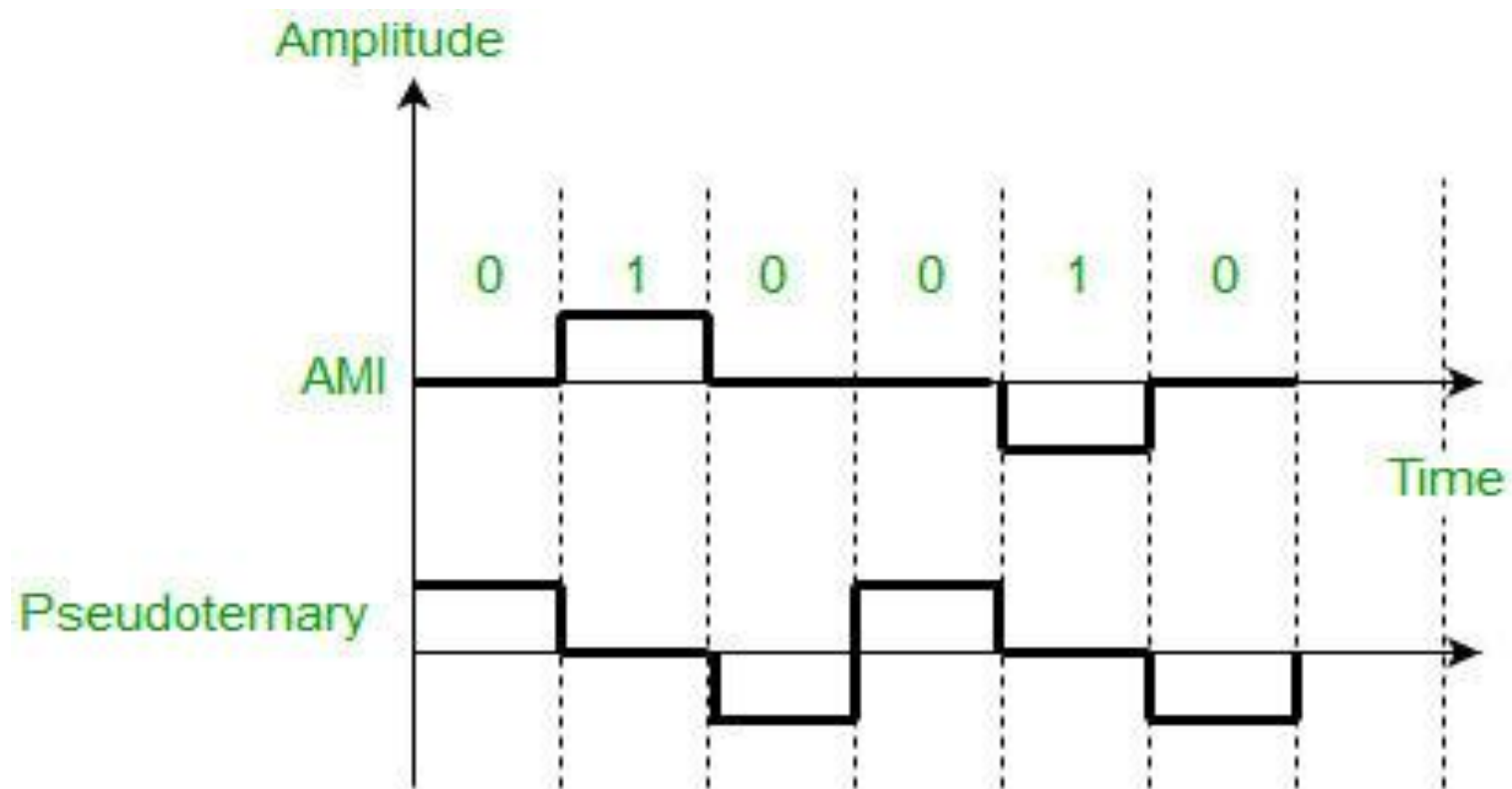
- $1 \rightarrow 0$
- $0 \rightarrow +V, -V$ alternately

Advantages:

- • No DC component

Drawbacks:

- • Synchronization issue



Bipolar Line Coding

Advantages:

- No DC component
- Better error detection capability
- Improved signal quality for long distances

Disadvantages:

- Long sequences of zero cause synchronization problems
- More complex encoding and decoding

Summary

- Unipolar: Simple but poor performance
- Polar: Better noise immunity
- Bipolar: No DC but complex